

Example 8.1.24: The convex hull of $\{0, 1\}$ in $\mathbb{R}$ is $[0, 1]$. Proof: A convex set containing 0 and 1 must contain $[0, 1]$, so $[0, 1] \subset \text{co}(\{0, 1\})$. The set $[0, 1]$ is convex and contains $\{0, 1\}$, so $\text{co}(\{0, 1\}) \subset [0, 1]$.

Linear mappings preserve convex sets. So in some sense, convex sets are the right sort of sets when considering linear mappings or changes of coordinates.

Proposition 8.1.25. Let X, Y be vector spaces, $A \in L(X, Y)$, and let $C \subset X$ be convex. Then $A(C)$ is convex.

Proof. Take two points $p, q \in A(C)$. Pick $u, v \in C$ such that $Au = p$ and $Av = q$. As C is convex, $(1 - t)u + tv \in C$ for all $t \in [0, 1]$. So

$$(1 - t)p + tq = (1 - t)Au + tAv = A((1 - t)u + tv) \in A(C). \quad \square$$

8.1.5 Exercises

Exercise 8.1.1: Show that in $\mathbb{R}^n$ (with the standard euclidean metric), for every $x \in \mathbb{R}^n$ and every $r > 0$, the ball $B(x, r)$ is convex.

Exercise 8.1.2: Verify that $\mathbb{R}^n$ is a vector space.

Exercise 8.1.3: Let X be a vector space. Prove that a finite set of vectors $\{x_1, x_2, \dots, x_n\} \subset X$ is linearly independent if and only if for every $k = 1, 2, \dots, n$

$$\text{span}(\{x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n\}) \subsetneq \text{span}(\{x_1, x_2, \dots, x_n\}).$$

That is, the span of the set with one vector removed is strictly smaller.

Exercise 8.1.4: Show that the set $X \subset C([0, 1], \mathbb{R})$ of those functions such that $\int_0^1 f = 0$ is a vector subspace. Compare [Exercise 8.1.16](#).

Exercise 8.1.5 (Challenging): Prove $C([0, 1], \mathbb{R})$ is an infinite-dimensional vector space where the operations are defined in the obvious way: $s = f + g$ and $m = af$ are defined as $s(x) := f(x) + g(x)$ and $m(x) := af(x)$. Hint: For the dimension, think of functions that are only nonzero on the interval $(1/(n+1), 1/n)$.

Exercise 8.1.6: Let $k: [0, 1]^2 \rightarrow \mathbb{R}$ be continuous. Show that $L: C([0, 1], \mathbb{R}) \rightarrow C([0, 1], \mathbb{R})$ defined by

$$Lf(y) := \int_0^1 k(x, y)f(x) dx$$

is a linear operator. That is, first show that L is well-defined by showing that Lf is continuous whenever f is, and then showing that L is linear.

Exercise 8.1.7: Let $\mathcal{P}_n$ be the vector space of polynomials in one variable of degree n or less. Show that $\mathcal{P}_n$ is a vector space of dimension $n + 1$.

Exercise 8.1.8: Let $\mathbb{R}[t]$ be the vector space of polynomials in one variable t . Let $D: \mathbb{R}[t] \rightarrow \mathbb{R}[t]$ be the derivative operator (derivative in t). Show that D is a linear operator.

Exercise 8.1.9: Let us show that [Proposition 8.1.18](#) only works in finite dimensions. Take the space of polynomials $\mathbb{R}[t]$ and define the operator $A: \mathbb{R}[t] \rightarrow \mathbb{R}[t]$ by $A(P(t)) := tP(t)$. Show that A is linear and one-to-one, but show that it is not onto.

Exercise 8.1.10: Finish the proof of [Proposition 8.1.17](#) in the finite-dimensional case. That is, suppose $\{x_1, x_2, \dots, x_n\}$ is a basis of X , $\{y_1, y_2, \dots, y_n\} \subset Y$, and define a function

$$A(x) := \sum_{k=1}^n b_k y_k, \quad \text{if } x = \sum_{k=1}^n b_k x_k.$$

Prove that $A: X \rightarrow Y$ is linear.

Exercise 8.1.11: Prove [Proposition 8.1.19](#). Hint: A linear transformation is determined by its action on a basis. So given two bases $\{x_1, \dots, x_n\}$ and $\{y_1, \dots, y_m\}$ for X and Y respectively, consider the linear operators A_{jk} that send $A_{jk}x_j = y_k$, and $A_{jk}x_\ell = 0$ if $\ell \neq j$.

Exercise 8.1.12 (Easy): Suppose X and Y are vector spaces and $A \in L(X, Y)$ is a linear mapping.

- Show that the nullspace $N := \{x \in X : Ax = 0\}$ is a vector space.
- Show that the range $R := \{y \in Y : Ax = y \text{ for some } x \in X\}$ is a vector space.

Exercise 8.1.13 (Easy): Show by example that a union of convex sets need not be convex.

Exercise 8.1.14: Compute the convex hull of the set of 3 points $\{(0, 0), (0, 1), (1, 1)\}$ in $\mathbb{R}^2$.

Exercise 8.1.15: Show that the set $\{(x, y) \in \mathbb{R}^2 : y > x^2\}$ is a convex set.

Exercise 8.1.16: Show that the set $X \subset C([0, 1], \mathbb{R})$ of those functions such that $\int_0^1 f = 1$ is a convex set, but not a vector subspace. Compare [Exercise 8.1.4](#).

Exercise 8.1.17: Show that every convex set in $\mathbb{R}^n$ is connected using the standard topology on $\mathbb{R}^n$.

Exercise 8.1.18: Suppose $K \subset \mathbb{R}^2$ is a convex set such that the only point of the form $(x, 0)$ in K is the point $(0, 0)$. Further suppose that $(0, 1) \in K$ and $(1, 1) \in K$. Show that if $(x, y) \in K$ and $x \neq 0$, then $y > 0$.

Exercise 8.1.19: Prove that an arbitrary intersection of vector subspaces is a vector subspace. That is, if X is a vector space and $\{V_\lambda\}_{\lambda \in I}$ is an arbitrary collection of vector subspaces of X , then $\bigcap_{\lambda \in I} V_\lambda$ is a vector subspace of X .

Exercise 8.1.20 (Easy): Finish the proof of [Proposition 8.1.16](#), that is, prove the first four items of the proposition.